18 2-bromopropane undergoes nucleophilic substitution at a faster rate than 2-chloropropane under the same conditions.

Which two statements explain this?

1. The ionisation energy of bromine is less than the ionisation energy of chlorine.
2. The shared pair of electrons in the carbon–halogen bond is more shielded from the halogen nucleus in 2-bromopropane than in 2-chloropropane.
3. The atomic radius of bromine is larger than the atomic radius of chlorine.
4. There are more protons in the nucleus of a bromine atom than in the nucleus of a chlorine atom.

A 1 and 3 **B** 2 and 3 **C** 2 and 4 **D** 1 and 4